

Equation Discovery for Extrapolation: Systematic Skeleton Search with a Projected Polynomial Background

Mika Ben Brückner

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1 Introduction

A central promise of machine learning is prediction *beyond* the observed data, and standard regressors are remarkably bad at it: a polynomial diverges at a rate dictated by its degree, a random forest predicts a constant outside the convex hull of the training inputs, and a multi-layer perceptron (MLP) extrapolates linearly. All three interpolate the data in Figure 1 almost perfectly, yet all three fail immediately outside the training domain. Symbolic regression (SR) offers a way out for an important class of problems: many systems in science and engineering can ultimately be described by comparatively simple closed-form equations, and if the data-generating process *is* such a formula f , recovering f up to symbolic equivalence yields exact extrapolation by construction. The price is a hard combinatorial search over expression space [Dong and Zhong, 2025], and when the data does *not* follow a formula expressible in the search language, SR has no extrapolation advantage (Section 4).

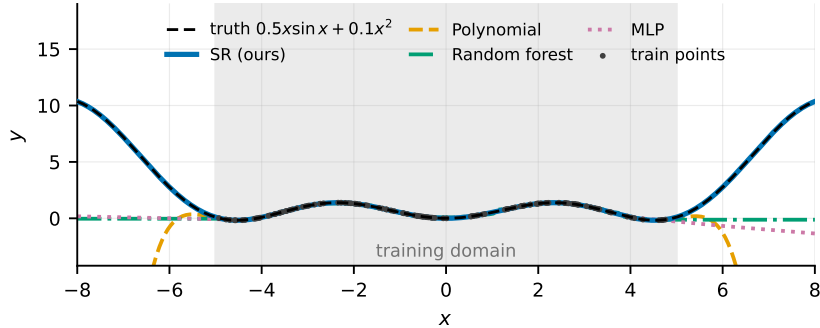


Figure 1: Extrapolation on $f(x) = 0.5x \sin x + 0.1x^2$ (128 noise-free training points on $[-5, 5]$). The baselines fit the training domain but fail outside it. Our method recovers the formula exactly ($\hat{f}(x) = 0.5x \sin x + 0.1x^2$) and extrapolates perfectly. Its curve coincides with the dashed ground truth.

Problem statement. Let $f : \mathbb{R}^d \rightarrow \mathbb{R}$ be an unknown formula over a known operator alphabet. We observe N_{train} samples (x_i, y_i) with $x_i \sim \text{Unif}([-a, a]^d)$ and $y_i = f(x_i) + \varepsilon_i$, $\varepsilon_i \sim \mathcal{N}(0, \sigma^2)$. The goal is a model $\hat{f}$ that minimizes the error on test points sampled *strictly outside* the training domain, $x \sim \text{Unif}([-b, b]^d \setminus [-a, a]^d)$ with $b > a$:

$$\text{MSE}_{\text{test}} = \frac{1}{N_{\text{test}}} \sum_{i=1}^{N_{\text{test}}} (\hat{f}(x_i) - f(x_i))^2. \quad (1)$$

This report describes (i) an SR algorithm combining an unbiased, complexity-ordered search over expression skeletons in the spirit of Kahlmeyer et al. [2025] with two extrapolation-specific components: a polynomial background solved in closed form by variable projection (generalizing linear scaling [Keijzer, 2003]), and model selection by *extrapolation-shaped* cross-validation, (ii) a synthetic generator that samples random formulas by a dice roll over operators, and (iii) experiments mapping out where the method beats polynomial, random-forest, and MLP baselines, and where it breaks.

2 Approach

Related work. SR systems search by genetic programming [Cranmer, 2023], by gradient descent through operator layers [Kim et al., 2021, Tsoi et al., 2025], or by continuous relaxation [Scholl et al., 2025], all biasing the search [Dong and Zhong, 2025] in ways that are hard to check [Kahlmeyer et al., 2025]. We adopt the assumption-free stance of Kahlmeyer et al. [2025], whose search is unbiased but randomized, and make it *systematic*: enumerate small expressions in order of size.

2.1 Model assumptions

Expression skeletons. We search over *skeletons*: expression trees whose internal nodes carry operators from $\mathcal{U} = \{\sin, \cos, \exp, \log, \sqrt{\cdot}, (\cdot)^2\}$ (unary) and $\mathcal{B} = \{+, -, \times, \div\}$ (binary), and whose leaves are either input variables x_j or constant placeholders c . A skeleton u with p placeholders defines a parametric family u_θ , $\theta \in \mathbb{R}^p$. Following Kahlmeyer et al. [2025], the only structural assumption we place on the search space is that it contains *small* expressions: the *core complexity* $k(u)$ is the number of operator nodes, and the search proceeds in order of increasing k . We use *protected* operators $\log|t|$ and $\sqrt{|t|}$ so that they impose no domain restrictions. Division is left unprotected, so skeletons using it may have poles, and invalid evaluations are penalized during fitting.

Regression model. A skeleton is turned into a regressor by attaching a linear *polynomial background*,

$$\hat{f}(x) = \alpha u_\theta(x) + \sum_{j=1}^m \beta_j m_j(x), \quad (2)$$

where $\{m_j\} = \{1\} \cup \{x_j^q : q \leq \deg\}$ (plus x_0x_1 for $d = 2$) are fixed monomials, $\deg \in \{2, 3\}$ by Section 3.2. The background, a generalization of linear scaling [Keijzer, 2003] from $\{1\}$ to a monomial basis, is the crucial ingredient: the core $x \sin x$ explains Figure 1 up to a polynomial trend, so with (2) it is the best of the 69 skeletons at $k = 2$, at machine precision. Under plain linear scaling it still ranks first but leaves 40% of the variance unexplained, and without the background it drops to rank 24 and is pruned. The *total complexity* of a fitted model is $C = k(u) + \#\{j \geq 2 : \beta_j \neq 0\}$, core operators plus active non-intercept background terms.

2.2 Parameter learning

For a fixed skeleton u the parameters split into a nonlinear part θ and a linear part (α, β) . We exploit this by *variable projection*: for given θ , the optimal linear coefficients are the ordinary least-squares solution

$$(\alpha^*, \beta^*)(\theta) = \arg \min_{\alpha, \beta} \|y - \alpha u_\theta(X) - M\beta\|_2^2, \quad M_{ij} = m_j(x_i), \quad (3)$$

computed in closed form. Substituting back leaves a nonlinear least-squares problem in θ only, $\min_\theta \|r(\theta)\|_2^2$ with $r(\theta) = y - \alpha^*(\theta) u_\theta(X) - M\beta^*(\theta)$, solved by a trust-region solver (finite-difference Jacobians) from multiple starts: $\theta^{(0)} = \mathbf{1}$ and one draw from $\mathcal{N}(0, 2^2 I)$ (two draws for $p \geq 3$). Non-finite predictions are mapped to a large constant to penalize invalid parameter regions. Skeletons with $p > 6$ placeholders are discarded. Selected models are finally *sparsified*: background terms with contribution $|\beta_j| \text{std}(m_j) < 10^{-4} \text{std}(y)$ are removed and the linear part refitted.

2.3 Search: exhaustive enumeration, then a diversity-aware beam

Semantically equivalent skeletons waste search budget, so trees are mapped to a canonical representative. Constant subtrees fold into a placeholder ($\sin(c) \rightarrow c$). Associative chains are flattened, sorted, and allowed at most one constant term. Parameter-absorbing compositions are rejected ($t - c$, t/c , e^{t+c} , $\log|ct|$, $\sqrt{|ct|}$, $(ct)^2$, $\log e^t$, $t + t$, $t \cdot t$, $t - t$, t/t). Operator aliases of the absolute value ($e^{\log|t|} = (\sqrt{|t|})^2 = |t|$) are kept only as $\sqrt{t^2}$. For $d = 1$ this leaves 1, 6, 69, 890, 12 769 skeletons with $k = 0, \dots, 4$ operator nodes.

These rules only remove duplicates they can *prove*. Many more skeletons are numerically redundant with respect to (2): since α and the intercept are free, cores agreeing up to an affine transform of their output are interchangeable, and with the full background so are cores differing by a background polynomial. We detect this with an *evaluation fingerprint*: each skeleton is evaluated on 24 fixed probe points for 3 generic parameter draws, the monomial basis is projected out (*background*) or not (*affine*), and the normalized result is hashed. A skeleton whose fingerprint was seen is skipped without fitting. The filter is heuristic in both directions: it can merge families that agree only on the probe points, and prune cores that are redundant alone but serve as intermediates. It is therefore validated in Section 3.2.

Algorithm 1 Complexity-ordered skeleton search and parameter learning. Both loops stop as soon as $\min \mathcal{F} < 10^{-10} \text{Var}(y)$.

Require: data (X, y) , 64-point subsample (X_s, y_s) , operators $(\mathcal{U}, \mathcal{B})$, limits $k_{\text{ex}}, k_{\text{max}}$, beam width B

- 1: $\mathcal{F} \leftarrow \emptyset$ $\triangleright$ front: best model per core complexity
- 2: **for** $k = 0, \dots, k_{\text{ex}}$ **do** $\triangleright$ Phase 1: exhaustive
- 3: $L_k \leftarrow$ all canonical skeletons with k operator nodes
- 4: FIT(u, X_s, y_s) for each $u \in L_k$; update $\mathcal{F}$
- 5: **end for**
- 6: **for** $k = k_{\text{ex}} + 1, \dots, k_{\text{max}}$ **do** $\triangleright$ Phase 2: beam
- 7: $P \leftarrow$ round-robin top- B of L_{k-1} by loss across root operators $\cup$ 5 random
- 8: $L_k \leftarrow$ canonical one-operator expansions of P (deduplicated)
- 9: FIT(u, X_s, y_s) for each $u \in L_k$; update $\mathcal{F}$
- 10: **end for**
- 11: refit $\mathcal{F}$ on all training points; sparsify; return Pareto-optimal $\mathcal{F}$
- 12: **procedure** FIT(u, X, y) $\triangleright$ variable projection, Section 2.2
- 13: $\theta^* \leftarrow \arg \min_{\theta} \|r(\theta)\|_2^2$ by multi-start trust-region NLS, where $r(\theta) = y - P(\theta)P(\theta)^+y$ and $P(\theta) = [u_{\theta}(X) \mid M]$
- 14: **return** θ^* , $(\alpha, \beta) = P(\theta^*)^+y$, loss $\|r(\theta^*)\|_2^2/N$
- 15: **end procedure**

Search runs in two phases (Algorithm 1). Phase 1 fits *every* canonical skeleton up to k_{ex} (3 or 4 for $d = 1$, 2 or 4 for $d = 2$). This is an unbiased search in the sense of Kahlmeyer et al. [2025], placing the method in the *deterministic* branch of the taxonomy of Dong and Zhong [2025], whose known trade-off (global optimality at exponentially growing cost) is exactly the behaviour we quantify in Section 3.3. Phase 2 grows skeletons one operator at a time with a beam. Expansions of a parent are all canonical trees obtained by replacing a leaf with a depth-one subtree or combining any node with a fresh leaf or unary wrapper. Ranking parents purely by training loss makes the beam greedy, so parents are chosen *round-robin across root-operator groups* in order of loss, plus 5 uniformly random survivors (ε -greedy). To bound cost, constants are fitted on a random subsample of 64 points during the search and refitted on all of them afterwards. The output is a *Pareto front* $\mathcal{F} = \{\hat{f}_k\}$: the best model per core complexity $k \leq k_{\text{max}}$, kept only where the training loss strictly improves.

2.4 Model selection: extrapolation-shaped cross-validation

The front $\mathcal{F}$ ranges from underfit to overfit models. Picking by training loss would always return the most complex one. Because our target is (1) rather than interpolation error, we validate the way we will be tested: training points are ranked by their sup-norm distance from the domain center, and for each quantile $q \in \{0.5, 0.6, 0.7, 0.8\}$ a fold is formed that refits each candidate’s constants on the inner q fraction and validates on the outer ring, giving cross-validation over *extrapolation horizons*. With $\text{cv}_{k,f}$ the validation MSE of $\hat{f}_k$ on fold $f \in F$, candidates are scored by $s_k = \frac{1}{|F|} \sum_f \log \text{cv}_{k,f}$, and the winner is chosen by the one-standard-error rule [Hastie et al., 2009, §7.10.1]: among all candidates with $s_k \leq s_{k^\dagger} + \widehat{\text{se}}(s_{k^\dagger})$, where $k^\dagger$ is the score minimizer and $\widehat{\text{se}}$ its standard error across folds, the one with the smallest total complexity C_k is selected and refitted on all training data. This makes the choice robust on training-loss plateaus, where CV differences between models are mostly noise. Candidates that diverge just outside the training box are removed beforehand by the growth guard motivated in Section 3.6. All hyperparameters of the baselines [Hastie et al., 2009, Ch. 3, 11, 15] are tuned on the *same* folds (Section 3.1). Ridge and the MLP run on standardized features, ours needs no scaling, so the comparison is fair.

2.5 Limits of the model

The hypothesis space is bounded by k_{max} (7 core operators for $d = 1$) and by the beam: formulas whose subterms only pay off jointly (e.g. the damped oscillation $e^{-c|x|} \cos(cx)$, $k = 7$) can be missed though representable. Protected log and $\sqrt{\cdot}$ extend past their natural domains [Keijzer, 2003]. Unprotected division admits poles that make fitting and evaluation ill-conditioned. Constant fitting is non-convex. Multi-starts mitigate but do not eliminate local minima (sin frequencies are the classic failure). Finally, the background biases the method towards “scaled core + polynomial” structure. Formulas like $\sin(x_0 x_1) \log |x_0|$ receive no help from it.

Table 1: Search configurations on 48 *hard* datasets ($d=1$, $\mathcal{O}_{\text{trig}}/\mathcal{O}_{\text{full}}$, 4–8 dice operators), varied one factor at a time around the default. Total time is a configuration’s cost over all 48 datasets. The last two rows arbitrate between all eleven configurations *per dataset* by the cross-validated score s_k of Section 2.4, while “oracle” instead picks the smallest test error and is an upper bound. Both report the selected run only.

Configuration	med. NMSE _{test}	Recovery	med. #fits	med. time [s]	total time [s]
default ($k_{\text{ex}}=3$, no dedup, $B=15$, deg 2)	$2.4 \cdot 10^{-30}$	67%	1783	5.2	5152
$k_{\text{ex}}=2$	$4.6 \cdot 10^{-30}$	62%	766	6.9	9601
$k_{\text{ex}}=4$	$5.0 \cdot 10^{-30}$	62%	2929	7.4	3182
dedup=affine	$3.1 \cdot 10^{-30}$	65%	1334	4.4	4952
dedup=background	$4.8 \cdot 10^{-30}$	69%	1280	6.3	5381
$k_{\text{ex}}=4$ +dedup	$8.6 \cdot 10^{-29}$	71%	1986	12.7	2838
beam=5	$9.4 \cdot 10^{-22}$	62%	1535	5.7	2942
beam=40	$2.4 \cdot 10^{-30}$	69%	2486	16.1	10797
poly_deg=0	$8.0 \cdot 10^{-29}$	62%	3282	53.8	5681
poly_deg=1	$3.1 \cdot 10^{-30}$	62%	2127	9.8	4792
poly_deg=3	$2.6 \cdot 10^{-30}$	71%	1364	6.8	3804
CV-selected per dataset	$2.4 \cdot 10^{-30}$	79%	656	2.6	3467
oracle (best test error)	$1.2 \cdot 10^{-30}$	79%	–	–	–

3 Experiments

3.1 Setup

Data generator. Ground truths are sampled by a dice roll: a random tree with a prescribed number of operator nodes over one of three alphabets ($\mathcal{O}_{\text{poly}}$: $\{+, -, \times, (\cdot)^2\}$; $\mathcal{O}_{\text{trig}}$: $\mathcal{O}_{\text{poly}} \cup \{\sin, \cos\}$; $\mathcal{O}_{\text{full}}$: all ten operators), with leaves x_j (probability 0.7) or constants from $\pm \text{Unif}([0.2, 3])$. A formula is rejected if, on a dense grid over $[-8, 8]^d$, it produces invalid values, $\max |f| > 10^4$ or $\text{std}(f) < 10^{-3}$, if it ignores a variable, or if it has ≥ 3 dice operators but is numerically affine. Because dice-rolled trees often simplify (e.g. $x(x-x-x)^2 = x^3$), analyses bin by the *effective complexity* k_{eff} , the operator count of the *sympy*-simplified formula. Datasets use $N_{\text{train}} = 128$, $N_{\text{test}} = 256$, $a = 5$, $b = 8$, and relative noise $\sigma = \sigma_{\text{rel}} \text{std}(f(X))$.

Metrics. We report the normalized test error $\text{NMSE} = \text{MSE}_{\text{test}}/\text{Var}(y_{\text{test}})$, aggregated by medians (and IQRs) since single blow-ups dominate means ($\text{NMSE} = 1 \approx$ predicting a constant), and the *recovery rate* [Kahlmeyer et al., 2025]: the fraction of runs whose model matches the ground truth on a dense grid over $[-8, 8]^d$ with relative RMSE below 10^{-6} , i.e. functional identity. Baselines are polynomial ridge regression (degree ≤ 8 , penalty on a grid), random forest (depth grid), and an MLP (width/penalty grid), all tuned on the extrapolation-shaped folds of Section 2.4. Everything is seeded, and the two commands in `README.md` reproduce all numbers.

3.2 Search configuration

We first fix the search hyperparameters by cross-validation, on the same extrapolation-shaped folds as Section 2.4 and never on test data. Table 1 reports a one-factor-at-a-time grid. Recovery differences between the levels of a factor never exceed four of the 48 datasets, so single rows indicate trends rather than resolved effects. Three observations drive the final configuration. *Deeper exhaustive enumeration pays off, but only together with deduplication*: $k_{\text{ex}} = 4$ alone recovers 62.5%, below the default, but with deduplication it reaches 70.8% versus 66.7% at 55% of the *total* compute, although its *median* run is slower. The extra level finds the formula in Phase 1 often enough to skip the beam, and with it the multi-minute worst cases that dominate the total. The *background* is confirmed as the decisive component: recovery is non-decreasing in its degree, and cutting it back to plain linear scaling [Keijzer, 2003] (deg=0) makes the median search ten times slower, because it then rarely terminates early. A *wider beam* ($B = 40$) buys almost nothing at double the cost, confirming that the bottleneck is the scoring of candidates rather than their number.

The CV-selected row of Table 1 arbitrates over all eleven configurations, not the deployed two. Since no single configuration dominates, the final method runs a small *portfolio* and keeps the model with the better cross-validated score. Two configurations, ($k_{\text{ex}}=4$, dedup) and (deg=3), already reach 79.2% (exactly the oracle rate of the full grid, at a ninth of its cost) because an exactly recovered formula drives its cross-validated error to numerical zero, a signal no incorrect configuration can imitate. All following experiments use this portfolio.

Table 2: Median $\text{NMSE}_{\text{test}}$ (lower is better, 10^{-30} is exact recovery, 1 is as bad as predicting a constant) and SR recovery rate, per operator alphabet and input dimension, from 25 ($d=1$) resp. 15 ($d=2$) random formulas per alphabet–complexity cell, noise-free. Note that $d=2$ omits $\mathcal{O}_{\text{poly}}$.

Setting	SR (ours)	Polynomial	Random forest	MLP	Recovery
d=1, poly	$8.2 \cdot 10^{-31}$	$6.6 \cdot 10^{-12}$	2.19	0.36	100%
d=1, trig	$7.1 \cdot 10^{-31}$	1.00	1.45	1.75	82%
d=1, full	$1.8 \cdot 10^{-30}$	1.07	1.25	0.96	73%
d=2, trig	$1.5 \cdot 10^{-31}$	0.25	0.74	0.15	89%
d=2, full	$1.2 \cdot 10^{-31}$	0.99	0.62	0.089	89%

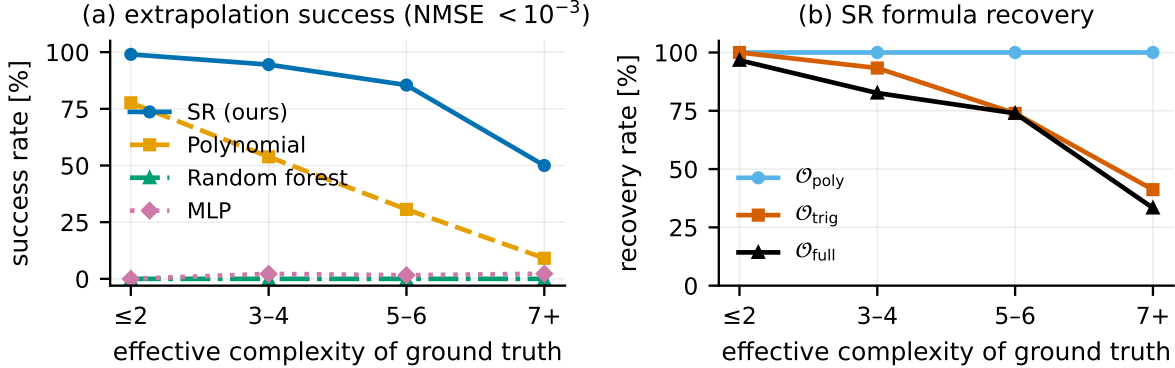


Figure 2: Main experiment ($d=1$, noise-free, 300 random formulas). (a) Fraction of runs with $\text{NMSE}_{\text{test}} < 10^{-3}$. (b) Fraction of runs where SR recovered the ground-truth formula exactly, per operator alphabet. Both panels are resolved by the effective complexity k_{eff} of the ground truth.

3.3 Extrapolation error and recovery

Table 2 and Figure 2 summarize the noise-free comparison. The median $\text{NMSE}_{\text{test}}$ of our method is at machine precision for *every* alphabet: in the majority of runs the formula is recovered exactly, and exact recovery implies exact extrapolation. Overall, 85% of the 300 ground truths are recovered (100% for $\mathcal{O}_{\text{poly}}$, 82% for $\mathcal{O}_{\text{trig}}$, 73% for $\mathcal{O}_{\text{full}}$), 88% of runs succeed in the sense of $\text{NMSE} < 10^{-3}$ and 90% reach $\text{NMSE} < 0.1$. The baselines confirm the failure modes from Section 1: the random forest *never* reaches $\text{NMSE} < 10^{-3}$ (it predicts a constant outside the training hull, median $\text{NMSE} \approx 1.2$ –2.2), the MLP succeeds in 1% of runs, and the polynomial succeeds in 51%, almost exactly the fraction of ground truths that are themselves polynomials of degree ≤ 8 , on which it is exact. On everything else it diverges (Table 2).

The aggregate hides the most characteristic property of the method: its error distribution is *bimodal*. The 75th percentile of $\text{NMSE}_{\text{test}}$ is still at machine precision while the 90th is already 0.13: the model either recovers the law or misses it, with almost nothing in between.

Figure 2 resolves this by ground-truth complexity: recovery decays from 99% at $k_{\text{eff}} \leq 2$ over 93% and 81% to 41% at $k_{\text{eff}} \geq 7$. This is the *complexity wall* of the bounded, partially greedy search, and precisely the exponential-cost trade-off that Dong and Zhong [2025] attribute to this method family. The wall is steeper for alphabets with log, exp, and division: these operators create poles and near-degenerate constants that both the dice roll and the search handle poorly. For $d=2$ (Table 2) recovery is 89% overall but drops from 100% at $k_{\text{eff}} \leq 4$ to 69% beyond, so the wall arrives one level earlier: the larger leaf set inflates every enumeration level. The baselines look less catastrophic in $d=2$ (MLP median NMSE down to 0.09) for a benign reason: most test-ring points are extreme in only *one* coordinate, so part of that task is near-interpolation. Runtimes are bimodal too, with a median of 0.03s against a worst case of 4.6min on a fruitless full-depth search, so the portfolio pays its $2.3\times$ overhead mostly on datasets it will fail on anyway.

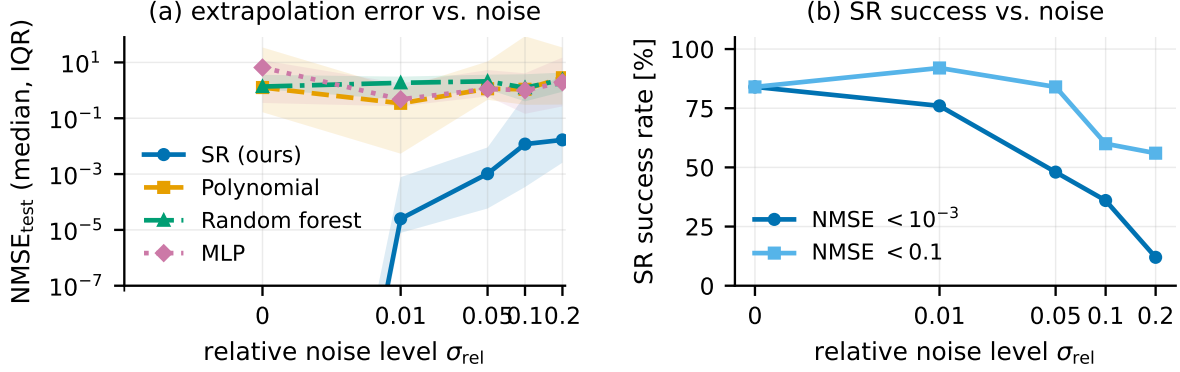


Figure 3: Noise experiment ($d=1$, $\mathcal{O}_{\text{full}}$, 25 formulas per level). (a) Median $\text{NMSE}_{\text{test}}$ (values below the axis are exact). (b) SR success rates at two thresholds.

Table 3: Selection ablation ($d=1$, $\mathcal{O}_{\text{full}}$, $k \in \{4, 6\}$ dice operators, 120 formulas per noise level): the same Pareto fronts, selected by extrapolation-shaped CV (ours), standard interior 3-fold CV, or raw training MSE. Differences are assessed by paired Wilcoxon signed-rank tests on log NMSE resp. C over the 120 shared datasets.

σ_{rel}	Selection	med. NMSE	p90 NMSE	Pr[NMSE > 1]	med. C
0	extrap-CV	$2.8 \cdot 10^{-30}$	1.14	0.11	4
0	interior-CV	$2.8 \cdot 10^{-30}$	1.21	0.12	4
0	train-MSE	$2.8 \cdot 10^{-30}$	1.76	0.12	4
0.1	extrap-CV	0.025	31.48	0.27	6
0.1	interior-CV	0.023	28.86	0.26	5
0.1	train-MSE	0.045	31.34	0.31	9

3.4 Noise

Figure 3 adds label noise $\sigma = \sigma_{\text{rel}} \text{std}(f(X))$. *Exact* recovery immediately drops to zero for any $\sigma > 0$, which is unavoidable since the fitted constants inherit an $O(\sigma)$ error, so the interesting quantity is the test error: SR’s median NMSE degrades gracefully from 0 over 10^{-3} ($\sigma_{\text{rel}} = 0.05$) to 0.017 at $\sigma_{\text{rel}} = 0.2$, while every baseline sits between 0.3 and 6.6 across all noise levels. At 20% noise our median error is still two orders of magnitude below the best baseline. The fraction of near-perfect extrapolations (NMSE < 10^{-3}) falls from 76% at $\sigma_{\text{rel}} = 0.01$ to 12% at 0.2, whereas the fraction of usable ones (NMSE < 0.1) only falls from 92% to 56%: noise therefore degrades the *precision* of the recovered law much faster than its *shape*.

3.5 Model-selection ablation

Table 3 isolates the selection step: each strategy picks from the *same* Pareto front, so any difference is due to the criterion alone. Noise-free, all three agree on almost every dataset. Under 10% noise two effects separate, of very different strength.

The *parsimony* effect is large and unambiguous: cross-validated selection never returns a more complex model than training-MSE selection and returns a strictly simpler one on 92 of 120 datasets (median $C = 6$ versus 9, $p \approx 6 \cdot 10^{-17}$). Its effect on accuracy is real but far more modest: the median error roughly halves (0.025 versus 0.045) and the blow-up rate falls from 0.31 to 0.27, which a paired test supports only weakly (52 wins, 40 losses, 28 ties, with $p = 0.043$).

The *fold geometry*, by contrast, has no measurable effect: extrapolation-shaped folds win on 23 datasets and lose on 24, with 73 exact ties ($p = 0.63$). The tie count explains the null result: in 61% of cases both geometries select the *identical* model, because the one-standard-error rule collapses the decision onto the complexity tie-break long before fold differences matter. We keep the extrapolation-shaped folds as the conceptually correct validation for the target regime, at no extra cost, but claim no empirical advantage for them. What earns its keep is parsimony.

3.6 Analysis of learned models

What the search finds. Recovery is judged *functionally*, and the search frequently returns a different but equivalent form: the dice-rolled $\sin(2x) - \sin(2x + 2.4)$ is found as $3.73 \cos(x) \sin(x - 0.37) + 0.68$ (a product-to-sum identity, $3.73 \approx 4 \sin 1.2$), and for the cubic truth $1.61 + 1.23(x - x^2)(1.08 - x)$ the core carries only x^3 and the background the rest. This is the division of labour intended by (2), visible in the learned models.

Failure taxonomy. The 30/300 runs with $\text{NMSE} > 0.1$ concentrate on three patterns: (i) *Poles*: truths like $1/(x - 1.32 + \cos x)$, whose test-domain behaviour is dominated by a singularity the training data barely samples. (ii) *Nested non-polynomial arguments*: log or exp of rational expressions ($\log(x - \log(x^2/1.46) - 2.05)$), which the beam only reaches through intermediate skeletons that fit poorly on their own. (iii) *Frequency fitting*: truths like $\sin(x + 2.83x^2)$, where the right skeleton is found but the nonconvex constant optimization misidentifies the frequency. Usually the selection then falls back to a simple, poorly fitting but *bounded* model rather than a wildly extrapolating one, a consequence of validating on extrapolation horizons.

A blind spot of in-domain validation, and its fix. Cross-validation has one structural blind spot. Run without the guard described below, the worst dataset of this study selected the core $\log(e^{-x} - e^{e^x})$ and reached $\text{NMSE} = 5 \cdot 10^{11}$, although that model is unremarkable on the training box (145 against $\max |y_{\text{train}}| = 145$). No in-domain fold can detect this, because extrapolation-shaped folds still only extrapolate to the edge of the data, never past it. We therefore add a *growth guard* before selection: candidates are evaluated on the training box dilated by 1.5, and those that are non-finite there or exceed $10^6 \max |y_{\text{train}}|$ are dropped (if all are, the front is kept). Both thresholds are deliberately loose (a legitimate e^x on $[-5, 5]$ grows only ≈ 12 -fold under this dilation), and no test data is used. The guard behaves exactly as a safety device should: comparing the 300-dataset grid with and without it, 299 selections are bit-identical and only the case above changes, its error dropping to $4 \cdot 10^{-5}$, so the worst case of the whole study falls to 153, at no cost in recovery.

Selection under noise. For the demo function with 5% noise the Pareto front collapses at $k = 2$ and then runs flat: every model above the elbow is the correct core plus a negligible decoration such as $+0.0088/\sqrt{|x|}$, bought with a training-MSE gain in the fifth decimal place. Such terms hardly affect $\text{NMSE}_{\text{test}}$ but destroy *symbolic* recovery, and no fit-based criterion rejects them. The one-standard-error rule selects the elbow, returning $0.504x \sin x + 0.102x^2 + 0.0006x - 0.013$.

4 Discussion

We built a symbolic regressor around three ideas: an unbiased search [Kahlmeyer et al., 2025] made systematic by complexity-ordered enumeration over canonical skeletons, a polynomial background whose coefficients are projected out in closed form so that partial explanations of the data are visible to the search early, and model selection that validates the way the model will be used, namely on extrapolation horizons, with a one-standard-error preference for simplicity. Its hyperparameters, including the choice between two search configurations per dataset, are set by cross-validation. On dice-rolled ground truths this recovers 85% of formulas exactly and beats the baselines by many orders of magnitude. Their failure is structural (constant or linear extrapolation, degree-driven divergence) and no tuning fixes it.

Not every component earned its keep equally. The polynomial background is decisive (recovery is non-decreasing in its degree, and cutting it back to plain linear scaling slows the search tenfold), and so is the parsimony rule. The extrapolation-shaped folds, however, are *not* measurably better than ordinary interior folds ($p = 0.63$): they cost nothing and are the conceptually right validation for the target regime, but the one-standard-error rule dominates the decision long before fold geometry matters.

The limits of Section 2.5 survive: deeper enumeration with deduplication pushed the complexity wall back but did not remove it, and even the final configuration recovers only 41% of the ground truths with $k_{\text{eff}} \geq 7$. More fundamentally, SR only pays off if the data-generating process *is* a formula over the chosen alphabet. Otherwise it degrades into an expensive, poorly-behaved regressor, and a black-box model with honestly constant extrapolation may be preferable. Natural extensions follow from these limits: the variable augmentation of Kahlmeyer et al. [2025] for divide and conquer, a core-term deletion test for Occam behaviour under noise, and neural guidance of the beam [Scholl et al., 2025].

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